

## KITTLA POKKA, Finland

WMO: 027170

Lat: 68.170N

Lon: 25.783E

Elev: 892

StdP: 14.23

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -26.5      | -20.6 | -32.9                         | 0.9 | -26.2 | -26.3 | 1.3 | -20.5 | 16.7                  | 22.3 | 15.2 | 21.6 | 3.6                   | 290  | 0.738 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 17.7                   | 76.3            | 60.9 | 71.7 | 59.2 | 67.6 | 57.0 | 63.3                | 71.6 | 61.1 | 69.0 | 58.9 | 65.2 | 6.9                  | 220  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 60.5                            | 81.3 | 65.9 | 58.1 | 74.4 | 62.8 | 56.1 | 69.3 | 60.7 | 29.2          | 71.4 | 27.6 | 69.2 | 26.0 | 65.3 | 70.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |          |          | Extreme Annual Temperature |          |                    |          | n-Year Return Period Values of Extreme Temperature |           |            |           |            |           |            |           |    |    |
|-------------------|----------|----------|----------------------------|----------|--------------------|----------|----------------------------------------------------|-----------|------------|-----------|------------|-----------|------------|-----------|----|----|
|                   |          |          | Mean                       |          | Standard Deviation |          | n=5 years                                          |           | n=10 years |           | n=20 years |           | n=50 years |           |    |    |
| 1%                | 2.5%     | 5%       | Min                        | Max      | Min                | Max      | Min                                                | Max       | Min        | Max       | Min        | Max       | Min        | Max       | DB | WB |
| (a) 14.7          | (b) 13.2 | (c) 11.7 | (e) -34.4                  | (f) 79.1 | (g) 4.8            | (h) 4.7  | (i) -37.8                                          | (j) 82.5  | (k) -40.7  | (l) 85.2  | (m) -43.4  | (n) 87.9  | (o) -46.9  | (p) 91.3  |    |    |
| (4) 14.7          | (5) 13.2 | (6) 11.7 | (7) -34.4                  | (8) 65.8 | (9) 4.6            | (10) 3.7 | (11) -37.7                                         | (12) 68.5 | (13) -40.4 | (14) 70.7 | (15) -43.0 | (16) 72.8 | (17) -46.3 | (18) 75.5 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 30.7   | 7.3   | 9.4   | 17.0  | 28.8  | 40.8  | 49.7  | 55.9  | 51.3  | 43.5  | 29.7  | 19.9  | 13.8  |  |  |
|                                                                        | DBStd       | 19.12  | 13.01 | 14.35 | 10.50 | 8.19  | 7.99  | 7.37  | 6.73  | 6.22  | 5.99  | 8.26  | 10.43 | 12.19 |  |  |
|                                                                        | HDD50       | 7458   | 1324  | 1138  | 1023  | 636   | 304   | 94    | 19    | 59    | 209   | 628   | 903   | 1121  |  |  |
|                                                                        | HDD65       | 12529  | 1789  | 1558  | 1488  | 1086  | 750   | 463   | 290   | 427   | 646   | 1093  | 1353  | 1586  |  |  |
|                                                                        | CDD50       | 418    | 0     | 0     | 0     | 0     | 19    | 84    | 203   | 99    | 13    | 0     | 0     | 0     |  |  |
|                                                                        | CDD65       | 14     | 0     | 0     | 0     | 0     | 0     | 3     | 9     | 2     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 223    | 0     | 0     | 0     | 0     | 12    | 35    | 150   | 26    | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH80       | 33     | 0     | 0     | 0     | 0     | 0     | 4     | 24    | 5     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | WSAvg       | 6.0    | 5.6   | 5.9   | 6.6   | 6.2   | 6.5   | 6.6   | 6.0   | 5.6   | 5.8   | 5.5   | 5.5   | 5.9   |  |  |
|                                                                        | PrecAvg     | 19.8   | 1.1   | 1.1   | 0.9   | 1.0   | 1.3   | 2.0   | 3.1   | 2.2   | 2.2   | 1.6   | 1.6   | 1.3   |  |  |
| (15) Precipitation                                                     | PrecMax     | 28.7   | 2.0   | 3.3   | 2.2   | 3.0   | 3.6   | 4.5   | 6.9   | 5.2   | 4.4   | 3.7   | 3.1   | 3.0   |  |  |
|                                                                        | PrecMin     | 14.7   | 0.2   | 0.2   | 0.0   | 0.2   | 0.2   | 0.0   | 0.7   | 0.6   | 0.3   | 0.1   | 0.3   | 0.2   |  |  |
|                                                                        | PrecStd     | 3.8    | 0.5   | 0.7   | 0.6   | 0.6   | 0.8   | 1.1   | 1.5   | 1.3   | 1.1   | 0.9   | 0.8   | 0.6   |  |  |
|                                                                        | PrecStd     | 3.8    | 0.5   | 0.7   | 0.6   | 0.6   | 0.8   | 1.1   | 1.5   | 1.3   | 1.1   | 0.9   | 0.8   | 0.6   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 32.2  | 36.5  | 39.4  | 53.8  | 74.7  | 78.5  | 83.3  | 77.0  | 62.3  | 48.3  | 40.9  | 35.2  |  |  |
|                                                                        |             | MCWB   | 31.7  | 34.8  | 34.7  | 43.9  | 55.9  | 61.4  | 63.6  | 65.3  | 55.4  | 45.0  | 40.6  | 33.7  |  |  |
|                                                                        | 2%          | DB     | 29.0  | 32.2  | 35.8  | 46.6  | 66.9  | 72.5  | 78.2  | 71.7  | 59.3  | 45.6  | 35.8  | 32.4  |  |  |
|                                                                        |             | MCWB   | 28.5  | 31.4  | 32.2  | 38.4  | 52.4  | 57.9  | 61.6  | 60.6  | 53.0  | 44.1  | 35.1  | 32.0  |  |  |
|                                                                        | 5%          | DB     | 26.7  | 30.3  | 33.3  | 42.9  | 60.2  | 67.5  | 73.7  | 66.7  | 56.7  | 43.2  | 33.3  | 31.0  |  |  |
|                                                                        |             | MCWB   | 26.3  | 29.6  | 31.0  | 36.8  | 48.5  | 55.0  | 60.1  | 57.7  | 51.5  | 42.1  | 33.0  | 30.7  |  |  |
|                                                                        | 10%         | DB     | 23.6  | 27.7  | 31.4  | 40.1  | 54.6  | 63.0  | 69.2  | 62.6  | 53.9  | 40.3  | 32.1  | 29.1  |  |  |
|                                                                        |             | MCWB   | 23.2  | 27.1  | 29.6  | 34.8  | 45.9  | 52.7  | 58.7  | 56.6  | 50.6  | 39.5  | 31.9  | 28.7  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 31.8  | 35.3  | 35.3  | 44.2  | 56.4  | 63.8  | 67.6  | 66.0  | 57.2  | 46.5  | 40.6  | 34.1  |  |  |
|                                                                        |             | MCDB   | 32.2  | 36.6  | 38.5  | 53.1  | 69.8  | 72.9  | 74.6  | 75.2  | 60.3  | 46.9  | 40.9  | 34.9  |  |  |
|                                                                        | 2%          | WB     | 28.5  | 31.7  | 33.1  | 39.7  | 53.3  | 60.5  | 63.9  | 62.0  | 54.7  | 44.4  | 34.9  | 32.2  |  |  |
|                                                                        |             | MCDB   | 28.8  | 32.2  | 35.0  | 45.1  | 66.0  | 68.9  | 72.2  | 68.3  | 57.4  | 45.5  | 35.4  | 32.5  |  |  |
|                                                                        | 5%          | WB     | 26.4  | 29.7  | 31.5  | 37.3  | 49.7  | 57.2  | 62.1  | 59.4  | 52.5  | 42.0  | 33.0  | 30.6  |  |  |
|                                                                        |             | MCDB   | 26.7  | 30.3  | 33.1  | 42.1  | 58.3  | 63.1  | 70.5  | 64.8  | 55.3  | 43.0  | 33.3  | 30.9  |  |  |
|                                                                        | 10%         | WB     | 23.3  | 27.2  | 29.7  | 35.6  | 47.0  | 54.4  | 60.3  | 57.3  | 50.7  | 39.7  | 32.0  | 28.7  |  |  |
|                                                                        |             | MCDB   | 23.6  | 27.8  | 31.3  | 39.3  | 54.0  | 61.3  | 68.0  | 61.2  | 53.3  | 40.4  | 32.1  | 29.1  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 14.7  | 14.5  | 18.6  | 16.9  | 16.7  | 16.6  | 17.7  | 17.3  | 13.9  | 10.2  | 11.4  | 13.4  |  |  |
|                                                                        |             | MCDBR  | 14.2  | 9.8   | 13.5  | 18.3  | 26.6  | 24.1  | 26.6  | 23.7  | 18.9  | 9.7   | 8.4   | 8.7   |  |  |
|                                                                        |             | MCWBR  | 14.3  | 9.4   | 11.1  | 11.9  | 14.2  | 11.5  | 12.7  | 13.7  | 12.7  | 8.3   | 8.4   | 8.7   |  |  |
|                                                                        | 5% WB       | MCDBR  | 14.1  | 9.8   | 11.2  | 16.5  | 23.5  | 20.6  | 21.5  | 19.6  | 14.3  | 9.2   | 8.3   | 8.8   |  |  |
|                                                                        |             | MCWBR  | 14.2  | 9.4   | 9.9   | 11.0  | 13.3  | 11.0  | 11.7  | 12.0  | 10.5  | 8.2   | 8.3   | 8.8   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.188 | 0.247 | 0.285 | 0.315 | 0.334 | 0.337 | 0.347 | 0.331 | 0.301 | 0.265 | N/A   | N/A   |  |  |
|                                                                        | taud        |        | 1.917 | 2.081 | 2.166 | 2.201 | 2.398 | 2.482 | 2.465 | 2.491 | 2.548 | 2.306 | N/A   | N/A   |  |  |
|                                                                        | Ebn at Noon |        | 30    | 172   | 236   | 263   | 272   | 275   | 267   | 259   | 240   | 170   | N/A   | 0     |  |  |
|                                                                        | Edh at Noon |        | 3     | 15    | 25    | 33    | 30    | 29    | 29    | 25    | 18    | 12    | N/A   | 0     |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 11    | 153   | 625   | 1231  | 1532  | 1566  | 1455  | 1045  | 550   | 181   | 28    | 0     |  |  |
|                                                                        | RadStd      |        | 1     | 17    | 38    | 70    | 150   | 124   | 149   | 87    | 56    | 31    | 3     | 0     |  |  |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (40 neighbors) | +1.06 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page